SG No. 175 C_{6h}^{1} $P6/m$ [hexagonal]

* plus set: $+[0,\ 0,\ 0]$

* Wyckoff site: `1a`, site symmetry: `6/m..`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `6/m..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,-7,-8,-9,-10,-11,-12]`

Table 2: Wyckoff bond: `3b@1a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ 0]$	`[1,-4,-7,10]`
2	$[-Y,\ X-Y,\ 0]$	$[0,\ 0,\ 0]$	`[2,-5,-8,11]`
3	$[-X+Y,\ -X,\ 0]$	$[0,\ 0,\ 0]$	`[3,-6,-9,12]`

Table 3: Wyckoff bond: `6c@1a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1,-7]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ 0]$	`[2,-8]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ 0]$	`[3,-9]`
4	$[-X,\ -Y,\ Z]$	$[0,\ 0,\ 0]$	`[4,-10]`
5	$[Y,\ -X+Y,\ Z]$	$[0,\ 0,\ 0]$	`[5,-11]`
6	$[X-Y,\ X,\ Z]$	$[0,\ 0,\ 0]$	`[6,-12]`

* Wyckoff site: `1b`, site symmetry: `6/m..`

Table 4: Wyckoff bond: `1a@1b`, bond symmetry: `6/m..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,-7,-8,-9,-10,-11,-12]`

Table 5: Wyckoff bond: `3b@1b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-4,-7,10]`
2	$[-Y,\ X-Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,-5,-8,11]`
3	$[-X+Y,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,-6,-9,12]`

Table 6: Wyckoff bond: `6c@1b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-7]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,-8]`

continued ...

Table 6

No.	vector	center	mapping
3	$[-X+Y,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,-9]`
4	$[-X,\ -Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,-10]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,-11]`
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[6,-12]`

* Wyckoff site: `2c`, site symmetry: `-6..`

Table 7: Wyckoff bond: `2a@2c`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[4,5,6,-7,-8,-9]`

Table 8: Wyckoff bond: `6b@2c`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,10]`
2	$[-Y,\ X-Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[2,11]`
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[3,12]`
4	$[-X,\ -Y,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[4,7]`
5	$[Y,\ -X+Y,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[5,8]`
6	$[X-Y,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[6,9]`

Table 9: Wyckoff bond: `12c@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[10]`
11	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[11]`
12	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[12]`

* Wyckoff site: `2d`, site symmetry: `-6..`

Table 10: Wyckoff bond: `2a@2d`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[4,5,6,-7,-8,-9]`

Table 11: Wyckoff bond: `6b@2d`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1,10]`
2	$[-Y,\ X-Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[2,11]`
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[3,12]`
4	$[-X,\ -Y,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[4,7]`
5	$[Y,\ -X+Y,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[5,8]`
6	$[X-Y,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[6,9]`

Table 12: Wyckoff bond: `12c@2d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[10]`
11	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[11]`
12	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[12]`

* Wyckoff site: `2e`, site symmetry: `6..`

Table 13: Wyckoff bond: `2a@2e`, bond symmetry: `6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,2,3,4,5,6]`
2	$[0,\ 0,\ -Z]$	$[0,\ 0,\ -z]$	`[7,8,9,10,11,12]`

Table 14: Wyckoff bond: `6b@2e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ z]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$[0,\ 0,\ z]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$[0,\ 0,\ z]$	`[3,-6]`

continued ...

Table 14

No.	vector	center	mapping
4	$[-X, -Y, 0]$	$[0, 0, -z]$	`[7,-10]`
5	$[Y, -X+Y, 0]$	$[0, 0, -z]$	`[8,-11]`
6	$[X-Y, X, 0]$	$[0, 0, -z]$	`[9,-12]`

Table 15: Wyckoff bond: `12c@2e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, z]$	`[3]`
4	$[-X, -Y, Z]$	$[0, 0, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, 0, z]$	`[5]`
6	$[X-Y, X, Z]$	$[0, 0, z]$	`[6]`
7	$[-X, -Y, -Z]$	$[0, 0, -z]$	`[7]`
8	$[Y, -X+Y, -Z]$	$[0, 0, -z]$	`[8]`
9	$[X-Y, X, -Z]$	$[0, 0, -z]$	`[9]`
10	$[X, Y, -Z]$	$[0, 0, -z]$	`[10]`
11	$[-Y, X-Y, -Z]$	$[0, 0, -z]$	`[11]`
12	$[-X+Y, -X, -Z]$	$[0, 0, -z]$	`[12]`

* Wyckoff site: `3f`, site symmetry: `2/m..`

Table 16: Wyckoff bond: `3a@3f`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-4,-7,10]`
2	$[-Y, X-Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-5,-8,11]`
3	$[-X+Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6,-9,12]`

Table 17: Wyckoff bond: `3b@3f`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,-7,-10]`
2	$[0, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,5,-8,-11]`
3	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,6,-9,-12]`

Table 18: Wyckoff bond: `6c@3f`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-7]`
2	$[-Y, X-Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-8]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-9]`
4	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-10]`

continued ...

Table 18

No.	vector	center	mapping
5	$[Y,\, -X+Y,\, Z]$	$\left[0,\, \frac{1}{2},\, 0\right]$	`[5,-11]`
6	$[X-Y,\, X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, 0\right]$	`[6,-12]`

* Wyckoff site: `3g`, site symmetry: `2/m..`

Table 19: Wyckoff bond: `3a@3g`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X,\, Y,\, 0]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[1,-4,-7,10]`
2	$[-Y,\, X-Y,\, 0]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,-5,-8,11]`
3	$[-X+Y,\, -X,\, 0]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,-6,-9,12]`

Table 20: Wyckoff bond: `3b@3g`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0,\, 0,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[1,4,-7,-10]`
2	$[0,\, 0,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,5,-8,-11]`
3	$[0,\, 0,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,6,-9,-12]`

Table 21: Wyckoff bond: `6c@3g`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[1,-7]`
2	$[-Y,\, X-Y,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,-8]`
3	$[-X+Y,\, -X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,-9]`
4	$[-X,\, -Y,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[4,-10]`
5	$[Y,\, -X+Y,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[5,-11]`
6	$[X-Y,\, X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[6,-12]`

* Wyckoff site: `4h`, site symmetry: `3..`

Table 22: Wyckoff bond: `4a@4h`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\, 0,\, Z]$	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	`[1,2,3]`
2	$[0,\, 0,\, Z]$	$\left[\frac{2}{3},\, \frac{1}{3},\, z\right]$	`[4,5,6]`
3	$[0,\, 0,\, -Z]$	$\left[\frac{2}{3},\, \frac{1}{3},\, -z\right]$	`[7,8,9]`
4	$[0,\, 0,\, -Z]$	$\left[\frac{1}{3},\, \frac{2}{3},\, -z\right]$	`[10,11,12]`

Table 23: Wyckoff bond: `12b@4h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	`[1]`

continued ...

Table 23

No.	vector	center	mapping
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[10]`
11	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[11]`
12	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[12]`

* Wyckoff site: `6i`, site symmetry: `2..`

Table 24: Wyckoff bond: `6a@6i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[3,-6]`
4	$[-X,\ -Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,-10]`
5	$[Y,\ -X+Y,\ 0]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[8,-11]`
6	$[X-Y,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,-12]`

Table 25: Wyckoff bond: `6b@6i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4]`
2	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[2,5]`
3	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[3,6]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,10]`
5	$[0,\ 0,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[8,11]`
6	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,12]`

Table 26: Wyckoff bond: `12c@6i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7]`

continued ...

Table 26

No.	vector	center	mapping
8	$[Y, -X+Y, -Z]$	$[0, \frac{1}{2}, -z]$	[8]
9	$[X-Y, X, -Z]$	$[\frac{1}{2}, \frac{1}{2}, -z]$	[9]
10	$[X, Y, -Z]$	$[\frac{1}{2}, 0, -z]$	[10]
11	$[-Y, X-Y, -Z]$	$[0, \frac{1}{2}, -z]$	[11]
12	$[-X+Y, -X, -Z]$	$[\frac{1}{2}, \frac{1}{2}, -z]$	[12]

* Wyckoff site: 6j, site symmetry: m..

Table 27: Wyckoff bond: 6a@6j, bond symmetry: m..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, y, 0]$	[1,10]
2	$[-Y, X-Y, 0]$	$[-y, x-y, 0]$	[2,11]
3	$[-X+Y, -X, 0]$	$[-x+y, -x, 0]$	[3,12]
4	$[-X, -Y, 0]$	$[-x, -y, 0]$	[4,7]
5	$[Y, -X+Y, 0]$	$[y, -x+y, 0]$	[5,8]
6	$[X-Y, X, 0]$	$[x-y, x, 0]$	[6,9]

Table 28: Wyckoff bond: 6b@6j, bond symmetry: m..

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, y, 0]$	[1,-10]
2	$[0, 0, Z]$	$[-y, x-y, 0]$	[2,-11]
3	$[0, 0, Z]$	$[-x+y, -x, 0]$	[3,-12]
4	$[0, 0, Z]$	$[-x, -y, 0]$	[4,-7]
5	$[0, 0, Z]$	$[y, -x+y, 0]$	[5,-8]
6	$[0, 0, Z]$	$[x-y, x, 0]$	[6,-9]

Table 29: Wyckoff bond: 12c@6j, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, 0]$	[1]
2	$[-Y, X-Y, Z]$	$[-y, x-y, 0]$	[2]
3	$[-X+Y, -X, Z]$	$[-x+y, -x, 0]$	[3]
4	$[-X, -Y, Z]$	$[-x, -y, 0]$	[4]
5	$[Y, -X+Y, Z]$	$[y, -x+y, 0]$	[5]
6	$[X-Y, X, Z]$	$[x-y, x, 0]$	[6]
7	$[-X, -Y, -Z]$	$[-x, -y, 0]$	[7]
8	$[Y, -X+Y, -Z]$	$[y, -x+y, 0]$	[8]
9	$[X-Y, X, -Z]$	$[x-y, x, 0]$	[9]
10	$[X, Y, -Z]$	$[x, y, 0]$	[10]
11	$[-Y, X-Y, -Z]$	$[-y, x-y, 0]$	[11]
12	$[-X+Y, -X, -Z]$	$[-x+y, -x, 0]$	[12]

* Wyckoff site: 6k, site symmetry: m..

Table 30: Wyckoff bond: `6a@6k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, y, \frac{1}{2}\right]$	`[1,10]`
2	$[-Y, X-Y, 0]$	$\left[-y, x-y, \frac{1}{2}\right]$	`[2,11]`
3	$[-X+Y, -X, 0]$	$\left[-x+y, -x, \frac{1}{2}\right]$	`[3,12]`
4	$[-X, -Y, 0]$	$\left[-x, -y, \frac{1}{2}\right]$	`[4,7]`
5	$[Y, -X+Y, 0]$	$\left[y, -x+y, \frac{1}{2}\right]$	`[5,8]`
6	$[X-Y, X, 0]$	$\left[x-y, x, \frac{1}{2}\right]$	`[6,9]`

Table 31: Wyckoff bond: `6b@6k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, y, \frac{1}{2}\right]$	`[1,-10]`
2	$[0, 0, Z]$	$\left[-y, x-y, \frac{1}{2}\right]$	`[2,-11]`
3	$[0, 0, Z]$	$\left[-x+y, -x, \frac{1}{2}\right]$	`[3,-12]`
4	$[0, 0, Z]$	$\left[-x, -y, \frac{1}{2}\right]$	`[4,-7]`
5	$[0, 0, Z]$	$\left[y, -x+y, \frac{1}{2}\right]$	`[5,-8]`
6	$[0, 0, Z]$	$\left[x-y, x, \frac{1}{2}\right]$	`[6,-9]`

Table 32: Wyckoff bond: `12c@6k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, y, \frac{1}{2}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[-y, x-y, \frac{1}{2}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[-x+y, -x, \frac{1}{2}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, -y, \frac{1}{2}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[y, -x+y, \frac{1}{2}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[x-y, x, \frac{1}{2}\right]$	`[6]`
7	$[-X, -Y, -Z]$	$\left[-x, -y, \frac{1}{2}\right]$	`[7]`
8	$[Y, -X+Y, -Z]$	$\left[y, -x+y, \frac{1}{2}\right]$	`[8]`
9	$[X-Y, X, -Z]$	$\left[x-y, x, \frac{1}{2}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x, y, \frac{1}{2}\right]$	`[10]`
11	$[-Y, X-Y, -Z]$	$\left[-y, x-y, \frac{1}{2}\right]$	`[11]`
12	$[-X+Y, -X, -Z]$	$\left[-x+y, -x, \frac{1}{2}\right]$	`[12]`

* Wyckoff site: `12l`, site symmetry: `1`

Table 33: Wyckoff bond: `12a@12l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-y, x-y, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x+y, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, -y, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[y, -x+y, z]$	`[5]`
6	$[X-Y, X, Z]$	$[x-y, x, z]$	`[6]`

continued ...

Table 33

No.	vector	center	mapping
7	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[7]
8	$[Y, -X+Y, -Z]$	$[y, -x+y, -z]$	[8]
9	$[X-Y, X, -Z]$	$[x-y, x, -z]$	[9]
10	$[X, Y, -Z]$	$[x, y, -z]$	[10]
11	$[-Y, X-Y, -Z]$	$[-y, x-y, -z]$	[11]
12	$[-X+Y, -X, -Z]$	$[-x+y, -x, -z]$	[12]